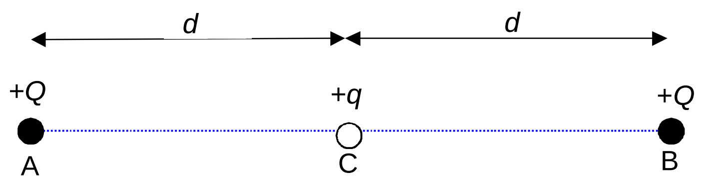
9	**(a)**	Define *electric potential*.		
		....		[2]
		The electric potential <u>at a point</u> in an electric field is defined as **the work done <u>per</u> unit <u>positive</u> charge by an <u>external agent</u> in moving a <u>point charge</u> <u>from infinity to that point</u>, (without any change in kinetic energy).**		
	(b)	Two identical point charges, of +*Q* each, are located at a distance 2*d* apart as shown in Fig 9.1.  **Fig. 9.1** A third charge of mass *m*, and charge +*q* is placed at C, midpoint between the two charges.		
		(i)	Obtain an expression for the electric potential at point C due to the two point charges at A and B.	
			electric potential at point C = ...V	[2]
			$V = V_A + V_B$ **$V = Q / (4\pi\varepsilon_o d) + Q / (4\pi\varepsilon_o d)$** **$= 2Q / (4\pi\varepsilon_o d)$** **$= Q / (2\pi\varepsilon_o d)$**	

(ii) Sketch in Fig. 9.2 to show how the electric potential varies with the position along AB for the following cases:

1. due to the point charge at A
2. due to the point charge at B
3. due to both point charges at A and B.

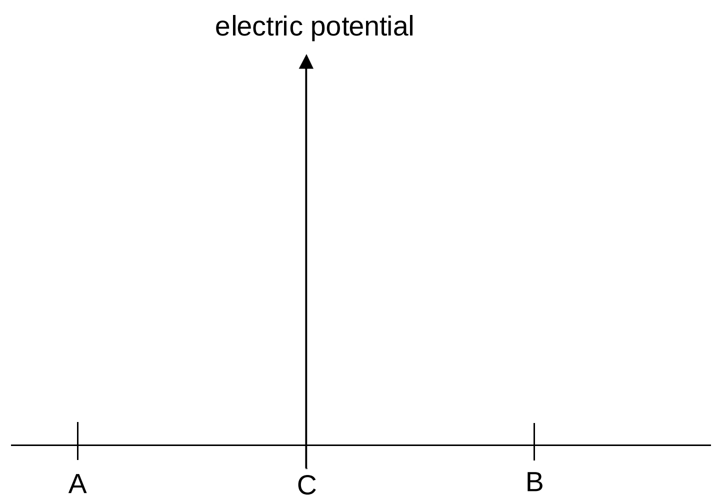
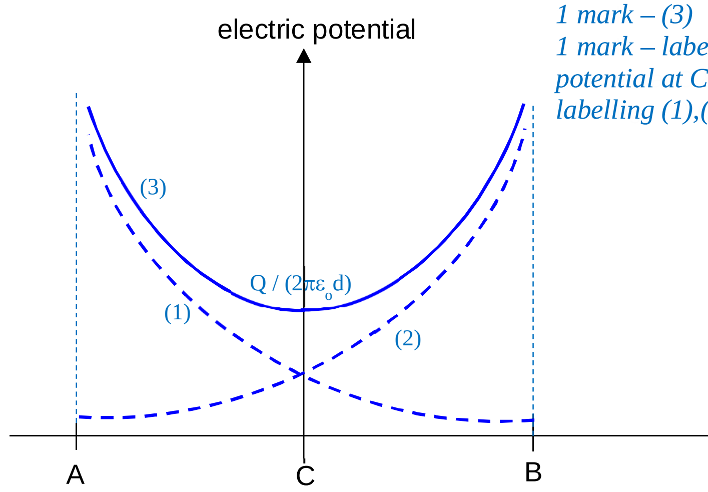


Fig. 9.2

[3]

Solution:

1 mark – (1) and (2)
1 mark – (3)
1 mark – labelling of potential at C & labelling (1),(2),(3)



(iii) The charge $+q$ is then being displaced horizontally and released. It is observed that charge $+q$ exhibits simple harmonic motion with an amplitude of x_o, where x_o is significantly smaller than d.

1. On Fig. 9.3, draw a well-labelled free body diagram to show the electrostatic forces acting on the third charge $+q$ at the instant when it is being displaced at a distance of x to the left of C.

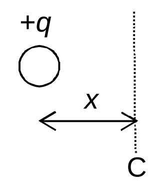
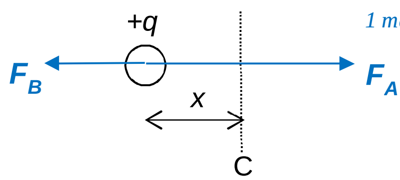


Fig. 9.3

[2]

Solution:

1 mark for label and directions
1 mark for length of arrows



2. Derive an expression for the magnitude of the resultant electrostatic force experienced by the charge due to the point charges at A and B.

resultant electrostatic force = ………………N [1]

Using $F = Q_1Q_2/(4\pi\varepsilon_o r^2)$

$F_R = F_A - F_B$

$= Qq\,/(4\pi\varepsilon_o(d-x)^2) - Qq\,/(4\pi\varepsilon_o(d+x)^2)$

$= \mathbf{qQ/(4\pi\varepsilon_o)\,[1/(d-x)^2 - 1/(d+x)^2]}$

3. The angular frequency of the oscillations can be expressed as follows.

$$\omega = \sqrt{\frac{qQ}{\pi\varepsilon_o m d^3}}$$

Determine the period of oscillation using the data provided below:

m: 6.8×10^{-25} kg
q: 3.2×10^{-19} C
Q: 1.28×10^{-18} C
d: 8.6 cm

period = ……………………… s [1]

Using $T = \frac{2\pi}{\omega}$

$$= 2\pi \times \sqrt{\frac{\pi\varepsilon_o \mathrm{m} d^3}{qQ}}$$

$$= 2\pi \times \sqrt{\frac{\pi 8.85 \mathrm{x} 10^{-12} \times 6.8 \mathrm{x} 10^{-25} \times 0.086^3}{3.2 \times 10^{-19} \times 1.28 \times 10^{-18}}}$$

$= 1.08$ s

4. On Fig. 9.4, sketch three graphs to show how the *total energy*, *potential energy* and k*inetic energy* of the charge +*q* vary as it oscillates between –x_o and x_o, Label each graph.

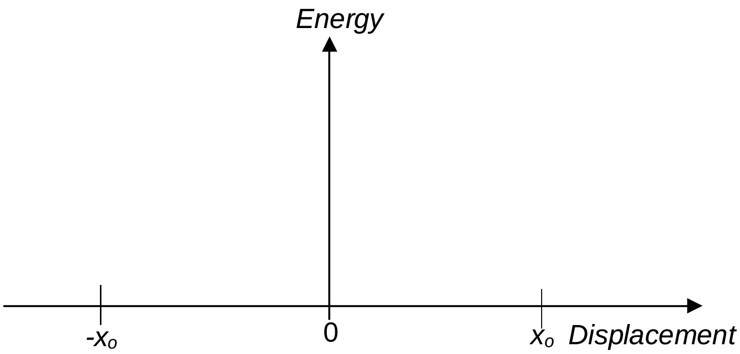
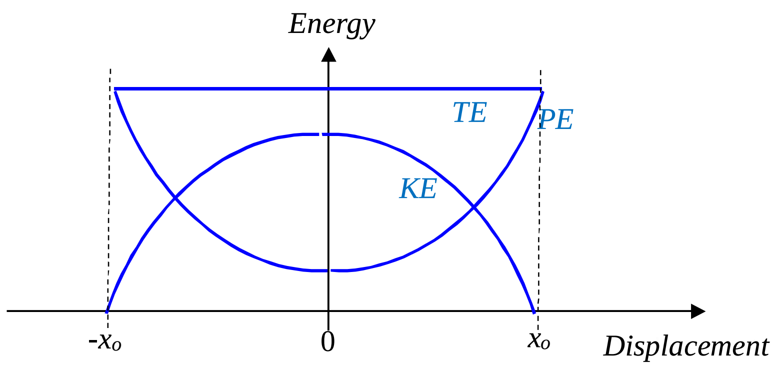


Fig. 9.4

[3]

Solution:



1 mark for each graph

(c) An electron was accelerated from rest with potential difference of 1.0 x 10^4 V, before entering at right angle into a uniform electric field region provided by a pair of charged parallel plates, separated at a distance of 3.0 cm as shown in Fig. 9.5.

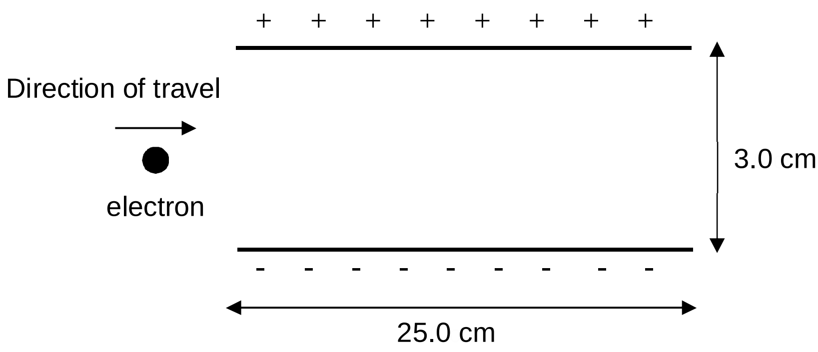


Fig. 9.5

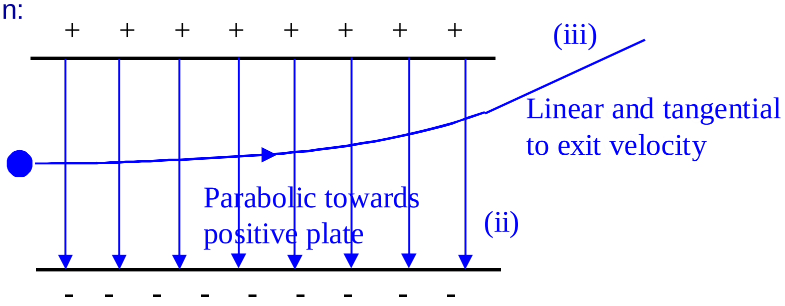
(i) Determine the speed of the electron when it entered the uniform electric field region.

speed = …………………… m s^{-1} [2]

During acceleration of the electron,
Gain in KE = Loss in EPE
$\frac{1}{2}mv^2 - 0 = q\,\Delta V$
$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = 1.60 \times 10^{-19} \times 1.0 \times 10^4$
$v = 5.9 \times 10^7$ m s^{-1}

(ii) On Fig. 9.5, draw the electric field lines between the pair of charged parallel plates. [1]

Solution:



(iii) The electron entered the uniform electric field region at the midpoint between the two charged parallel plates.

1. On Fig. 9.5, sketch the trajectory of the electron within and beyond the parallel plates. [1]

2. Show that the electron did not hit any of the parallel plates when it emerged from the uniform electric field region. The electric field strength between the parallel plates is 5.7×10^3 N C^{-1}.

[2]

				Horizontally: $s_x = u_x t$ $0.25 = 5.9 \times 10^7 \times t$ $t = 4.2 \times 10^{-9}$ s **Vertically:** $s_y = u_y t + \frac{1}{2} a_y t^2$ $= \frac{1}{2} \times (qE/m) \times t^2$ $= \frac{1}{2} \times (1.6\times10^{-19} \times 5.7\times10^3 / (9.11\times10^{-31})) \times (4.2\times10^{-9})^2$ $= 0.0088 \text{ m} < (\frac{1}{2} \times 0.030) \text{ m}$	